

Inverse Functions (Homework)

Name
Date
Course

Relations are written below in table form. Some of the tables could represent functions that are one-to-one, some of them could represent functions that are not one-to-one, and some cannot represent functions. For each table, write "1-to-1 function," "function," or "not a function."

①

x	16	-5	11	12	-2
y	19	-4	10	-1	-6

②

x	26	13	-2	6	7
y	-10	80	14	4	4

③

x	0	3	2	1	3
y	100	110	120	130	140

④

x	-9	3	-9	-3	-9
y	-1	-2	-3	-3	7

⑤

x	3	36	40	26	18
y	-3	36	40	16	15

⑥

x	0	2	0	8	9
y	3	7	4	7	3

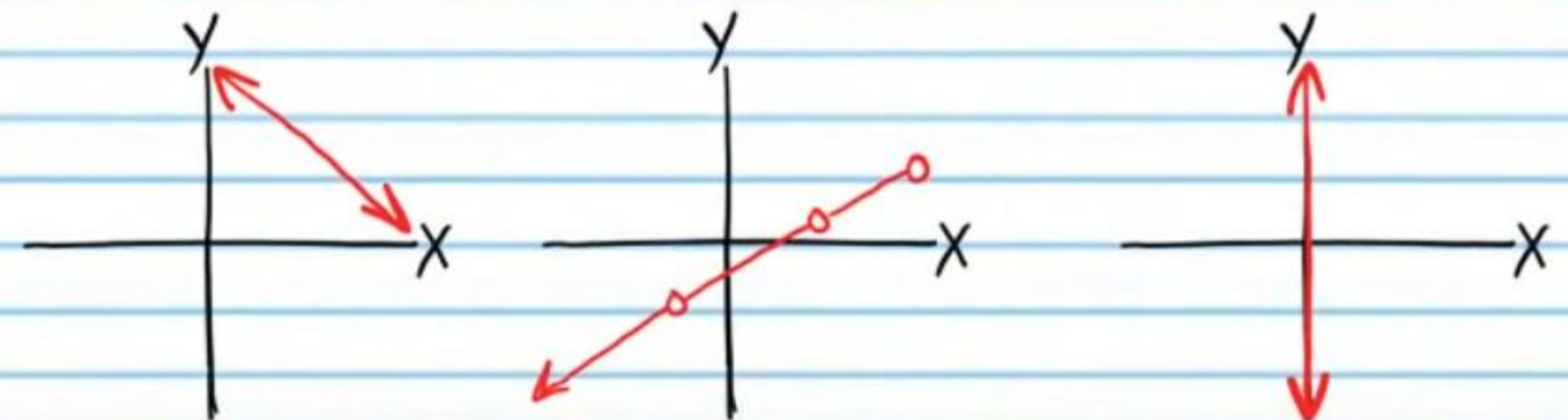
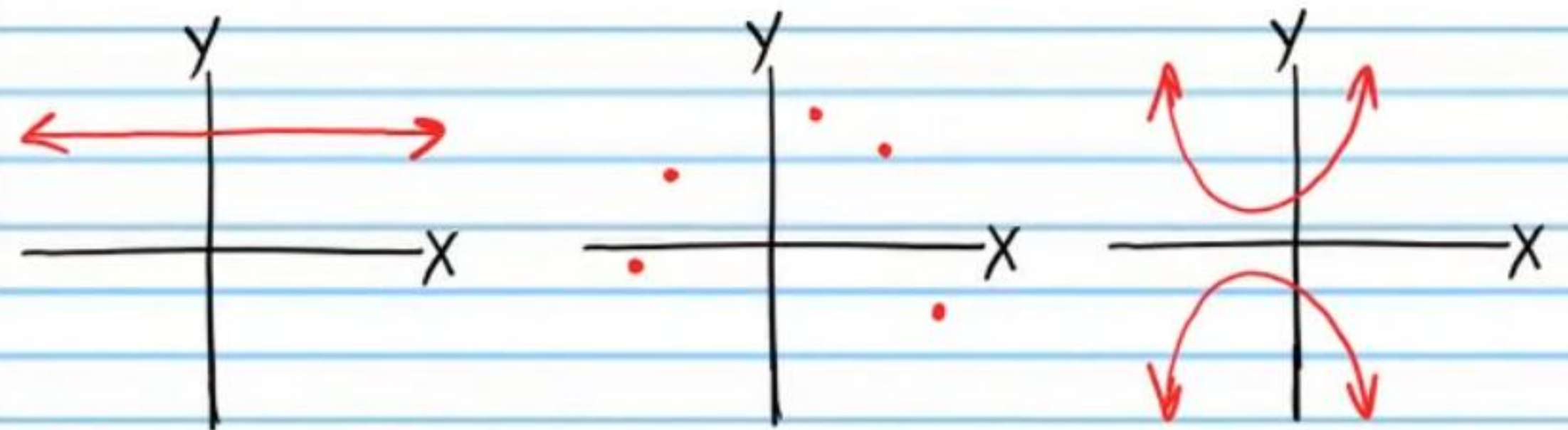
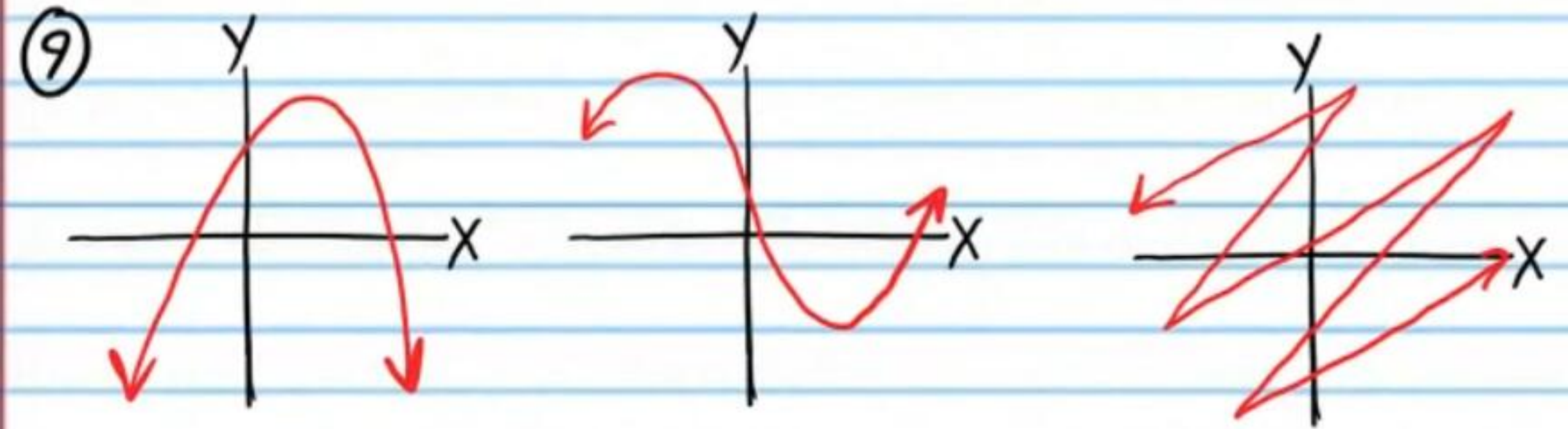
⑦

x	10	9	8	6	0
y	13	-8	$\frac{3}{4}$	$\frac{3}{4}$	-7

⑧

x	-17	4	-2	24	23
y	20	5	-9	-6	7

Relations are written below in the form of graphs. Some of the graphs could represent functions that are one-to-one, some of them could represent functions that are not one-to-one, and some cannot represent functions. Use the Vertical Line Test and the Horizontal Line Test to determine the scenario for each graph. Write "1-to-1 function," "function," or "not a function."



Each table below represents a function "f". Write the values of the inverse function "f⁻¹". If "f" is not a one-to-one function, then you must omit values of "f⁻¹" so that it qualifies as a function. To ensure that the functions are perfectly matched inverses, values of "f" may need to be omitted also. You may choose which values to omit. Highlight the omitted values.

⑩

X	f(x)	X	f ⁻¹ (x)
56	49		
50	50		
60	61		
59	58		
63	60		
61	59		

⑪

X	f(x)	X	f ⁻¹ (x)
4	-12		
7	-7		
-8	-12		
0	8		
14	14		
15	15		

⑫

X	f(x)	X	f ⁻¹ (x)
-2	5		
2	5		
4	1		
-2	4		
0	-1		
3	5		

⑬

X	f(x)	X	f ⁻¹ (x)
0	1		
1	2		
2	3		
3	4		
4	5		
5	6		

If "f" and "g" are inverse functions, answer the following questions.

⑭ If $f(3) = 4$, find $g(4)$.

⑮ If $g(-0.2) = -9$, find $f(-9)$.

⑯ If $f(0) = -\frac{3}{2}$, find $g(-\frac{3}{2})$.

⑰ If $g(-6) = 18$, find $f(18)$.

In the following problems, "f" and "f⁻¹" are inverse functions.

⑱ If $f(-10) = -8$, find $f^{-1}(-8)$.

⑲ If $f(-1) = 7.101$, find $f^{-1}(7.101)$.

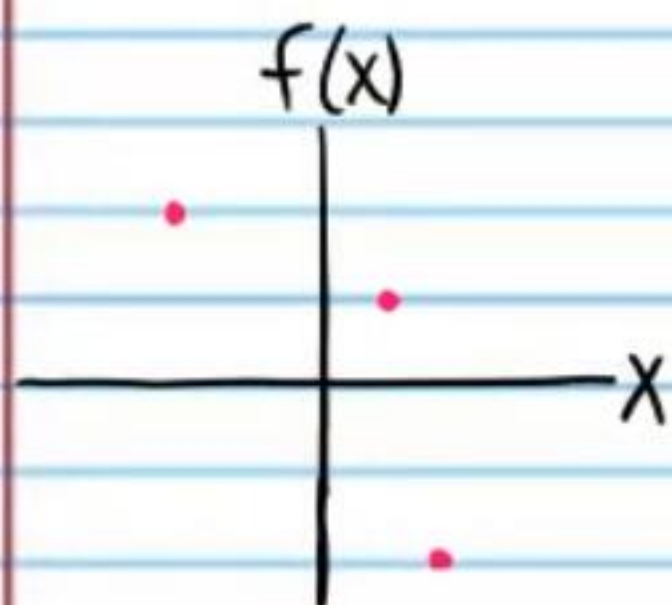
⑳ If $f^{-1}(\frac{1}{2}) = -\frac{1}{2}$, find $f(-\frac{1}{2})$.

㉑ If $f^{-1}(80) = 79$, find $f(79)$.

Each graph below represents a function "f." Draw the graph of the inverse function "f⁻¹." If "f" is not a one-to-one function, then you must erase parts of the graph of f⁻¹ so that it qualifies as a function. You may also highlight parts of the original graph that need to be erased so that the two inverses are perfectly matched. Use the Horizontal Line Test and the Vertical Line Test.

(22)

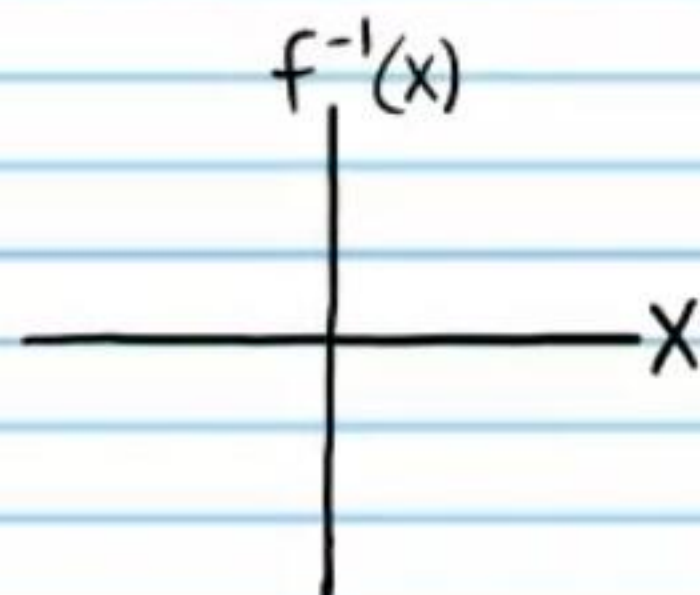
Write the inverse function
i.e. Reflection across $y=x$



Step 1
Reflect across the y-axis

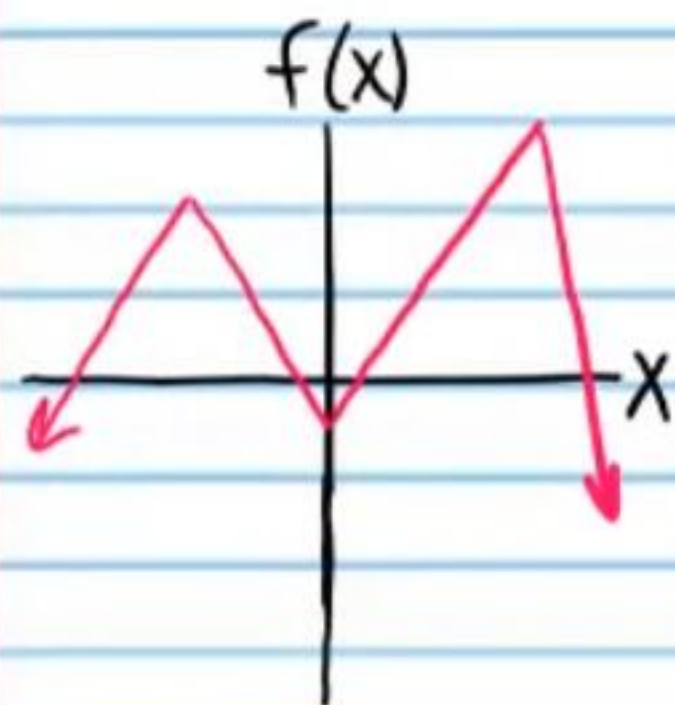


Step 2
Rotate 90° clockwise

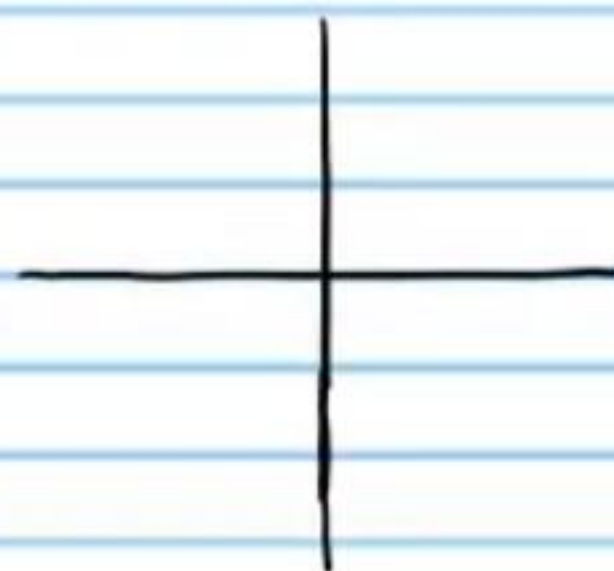


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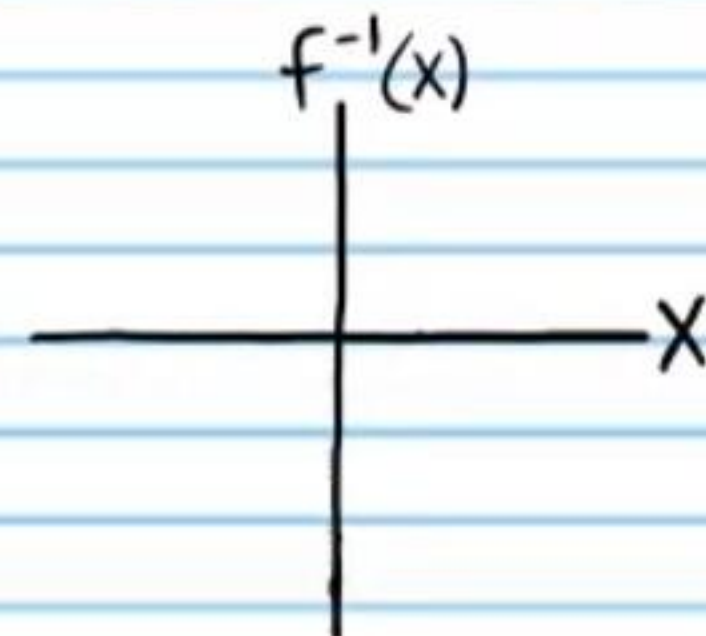
Write the inverse function
i.e. Reflection across $y=x$



Step 1
Reflect across the y-axis

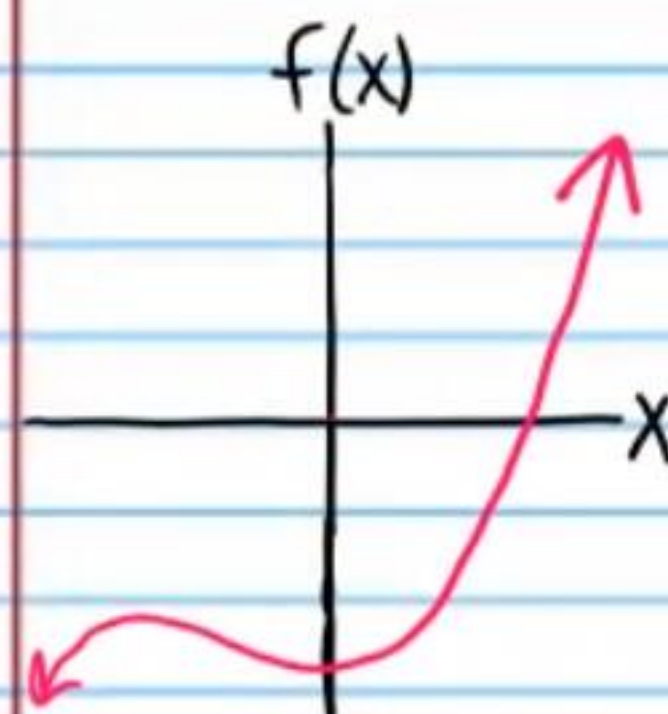


Step 2
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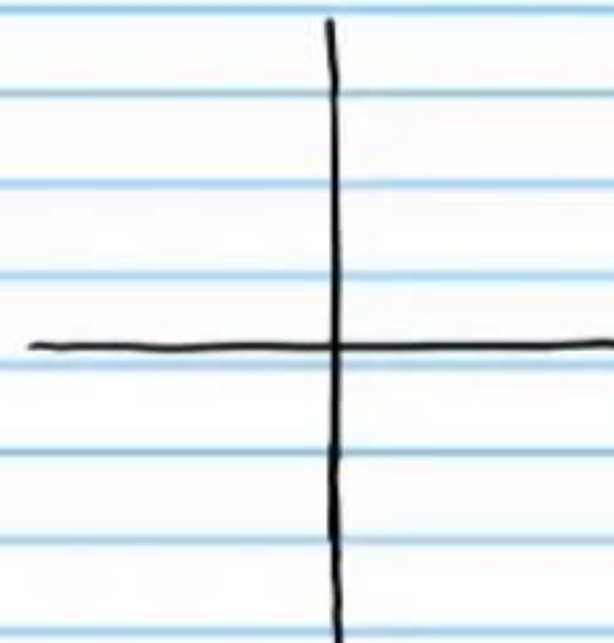


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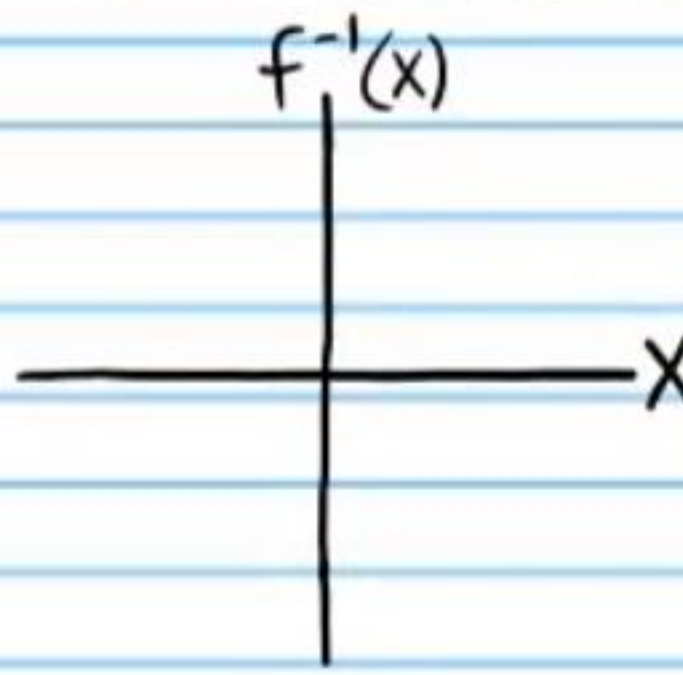
Write the inverse function
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Step 1
Reflect across the y-axis

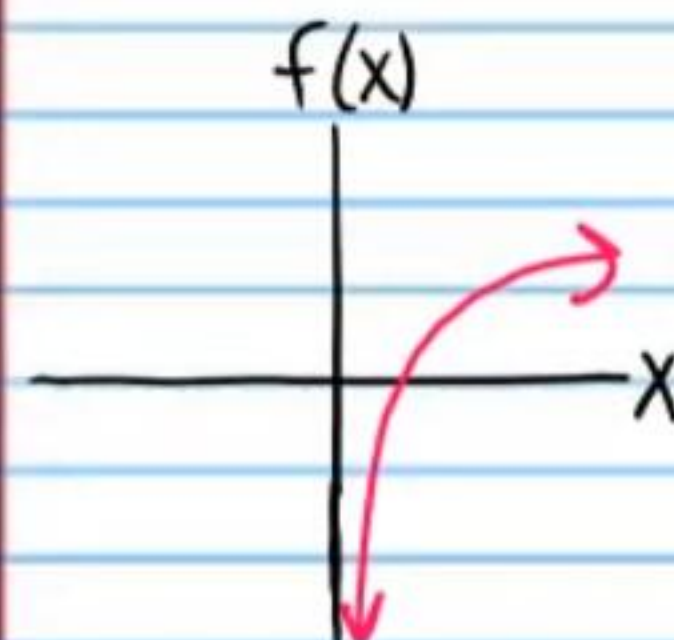


Step 2
Rotate 90° clockwise

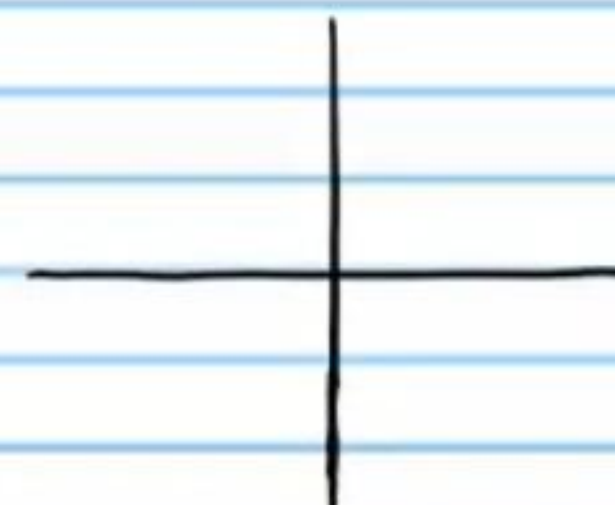


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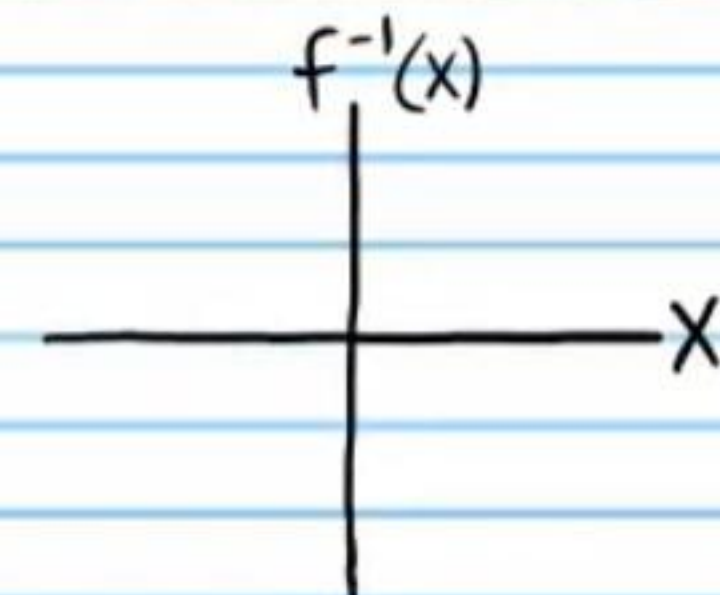
Write the inverse function
i.e. Reflection across $y=x$



Step 1
Reflect across the y-axis



Step 2
Rotate 90° clockwise



Find the equation of the inverse function.

(26) $f(x) = \frac{x-2}{5} + 1$ (27) $f(x) = x^7 + 3$

(28) $f(x) = -2 - 5x$ (29) $f(x) = \frac{1}{2}(x+2) - 8$

$$\textcircled{30} f(x) = 4^x$$

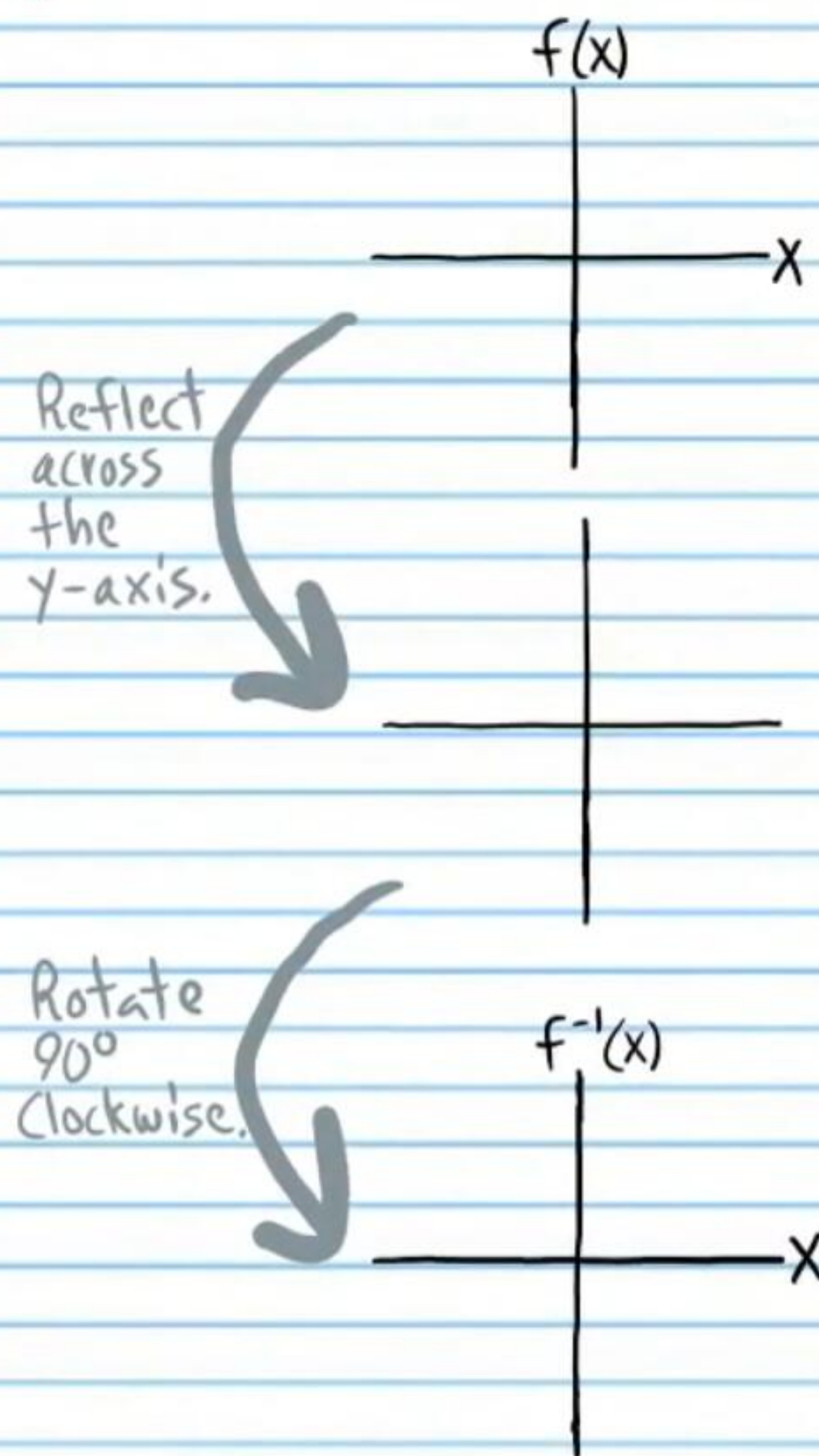
$$\textcircled{31} f(x) = 15^x$$

$$\textcircled{32} f(x) = \log_{26} x$$

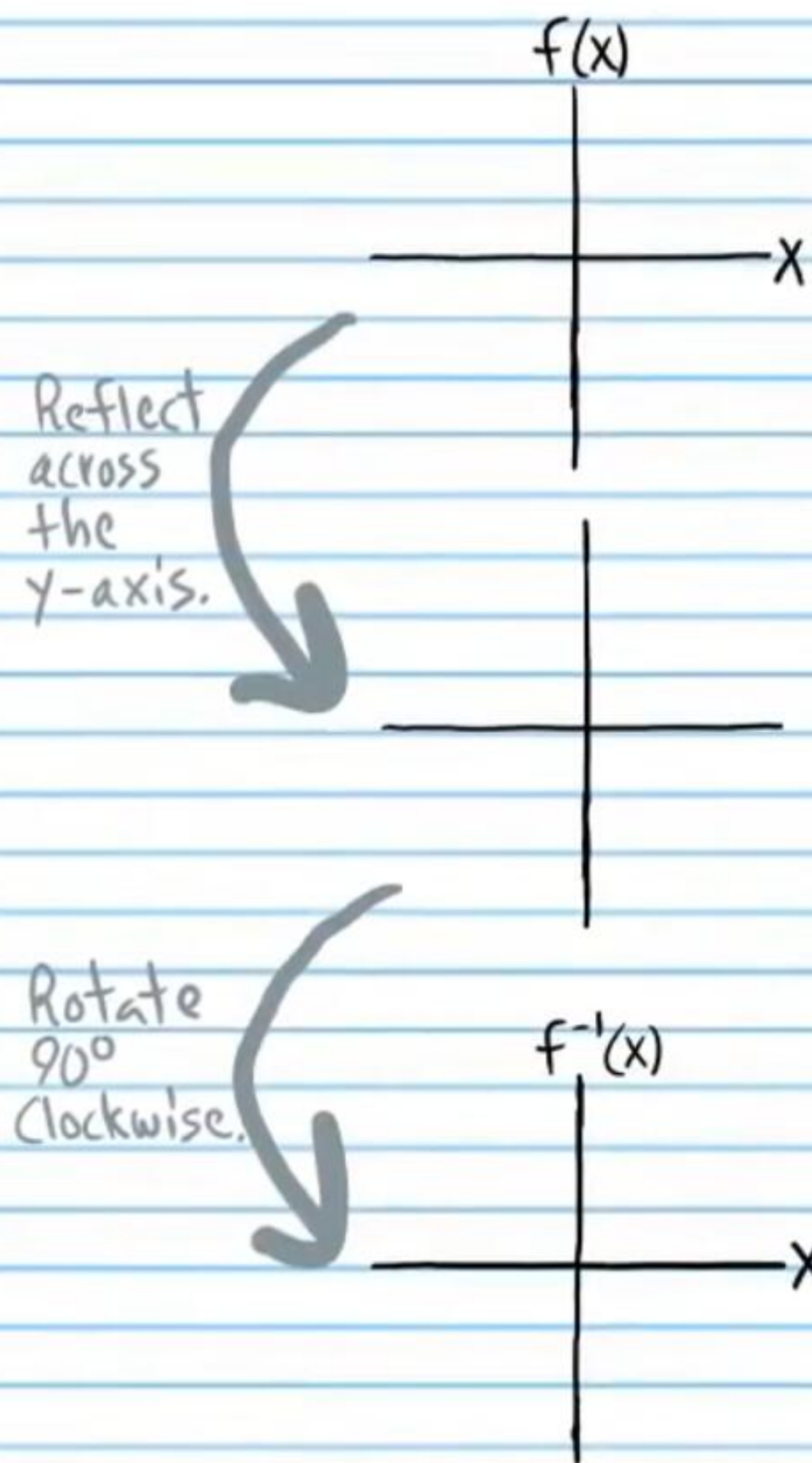
$$\textcircled{33} f(x) = \log_7 x$$

In the following problems, restrictions must be used on the original function and the inverse function so that they are perfectly matched and no aberrations occur. To understand this, graph the function and note that it fails the Horizontal Line Test, which means that it is not one-to-one. Draw the inverse graph and note that it fails the Vertical Line Test. Highlight the parts of the inverse graph and the original graph that you will omit and use inequality notation next to the equations to indicate restrictions.

$$\textcircled{34} f(x) = x^2$$



③⑤ $f(x) = \sqrt{x} - 3$



Homework Answers

① 1-to-1 Function

② Function

③ Not a Function

④ Not a Function

⑤ 1-to-1 Function

⑥ Not a Function

⑦ Function

⑧ 1-to-1 Function

⑨ Function

Function

Not a Function

Function

1-to-1 Function

Not a Function

1-to-1 Function

1-to-1 Function

Not a Function

⑩

X	$f^{-1}(x)$
49	56
50	50
61	60
58	59
60	63
59	61

⑪

X	$f^{-1}(x)$
-12	4
-7	7
-12	-8
8	0
14	14
15	15

⑫

X	$f^{-1}(x)$
5	-2
5	2
1	4
4	-2
-1	0
5	3

⑬

X	$f^{-1}(x)$
1	0
2	1
3	2
4	3
5	4
6	5

⑭ 3 ⑮ -0.2 ⑯ 0 ⑰ -6 ⑱ -10 ⑲ -1

⑳ $\frac{1}{2}$

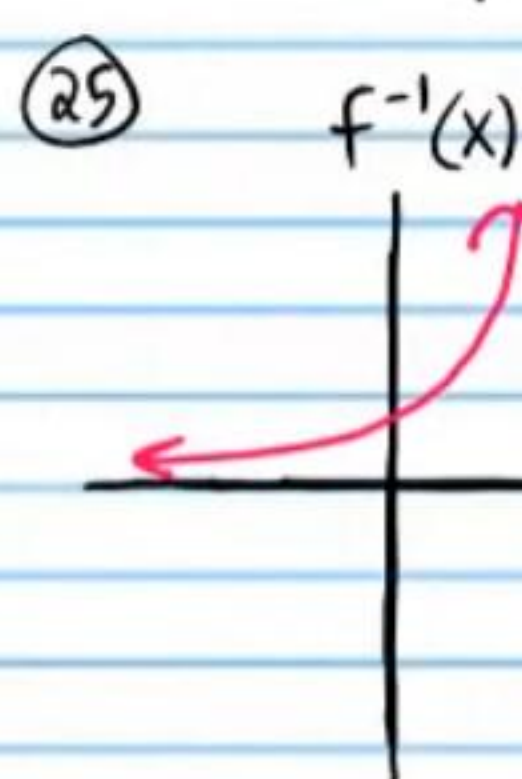
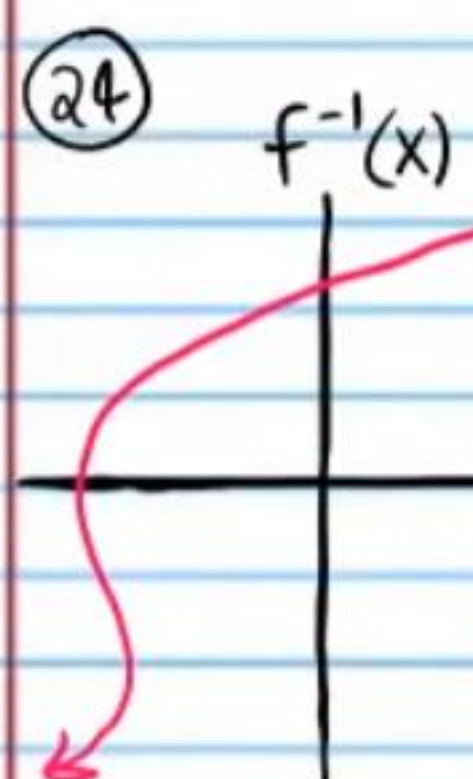
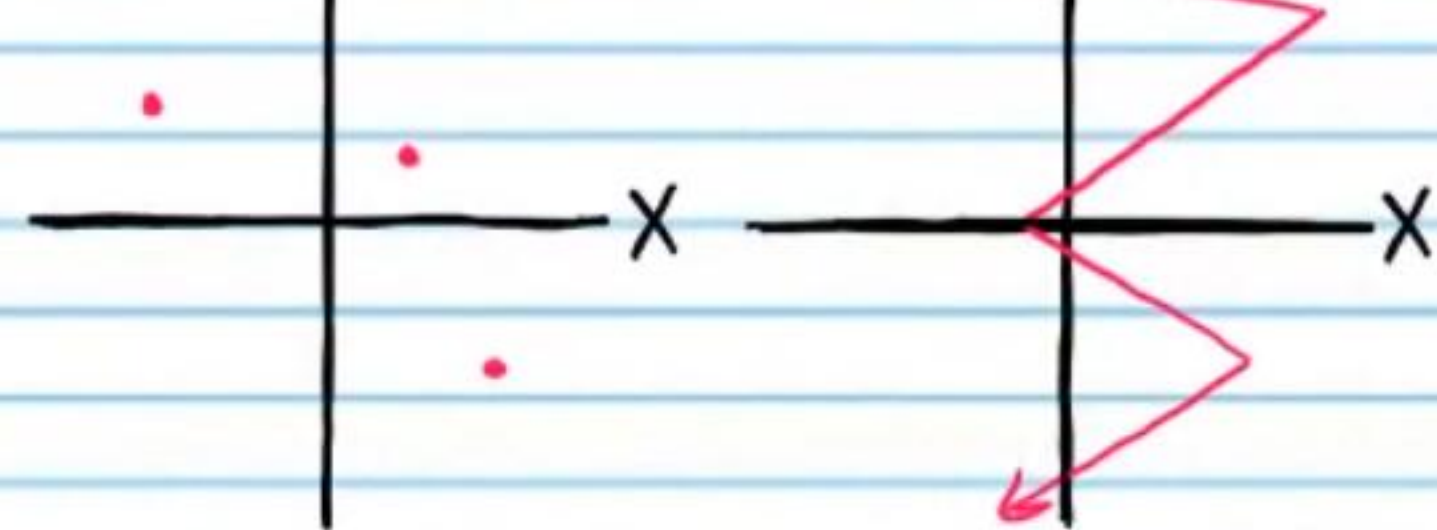
㉑ 80

㉒

$f^{-1}(x)$

㉓

$f^{-1}(x)$



㉖ $f^{-1}(x) = 5(x-1) + 2$
or
 $f^{-1}(x) = 5x - 3$

㉗ $f^{-1}(x) = \sqrt{x-3}$

㉙ $f^{-1}(x) = 2(x+8) - 2$

㉘ $f^{-1}(x) = -\frac{x+2}{5}$

or
 $f^{-1}(x) = 2x + 14$

$$(30) \quad f^{-1}(x) = \log_4 X$$

$$(31) \quad f^{-1}(x) = \log_{15} X$$

$$(32) \quad f^{-1}(x) = 26^x$$

$$(33) \quad f^{-1}(x) = 7^x$$

$$(34) \quad f(x) = x^2, \quad x \geq 0 \quad \text{or} \quad f(x) = x^2, \quad x \leq 0$$

$$f^{-1}(x) = \sqrt{x}$$

$$f^{-1}(x) = -\sqrt{x}$$

$$(35) \quad f^{-1}(x) = (x+3)^2, \quad x \geq -3$$